01 — Project Charter

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

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1. Problem Statement

The human genome is approximately 3.2 billion base pairs long, yet only about 1.5–2% encodes proteins. The remaining ~98% — the non-coding genome — contains regulatory elements (enhancers, promoters, silencers, insulators) that control when, where, and how genes are expressed. Many disease-associated genetic variants identified by genome-wide association studies (GWAS) fall in non-coding regions, yet we lack systematic, interpretable methods to understand which non-coding sequences act as regulatory "switches" and why.

Current computational approaches face a tension:

- **High-capacity models** (large genomic transformers, foundation models) can achieve strong predictive accuracy but require substantial computational resources — often hundreds of GPU-hours and large institutional infrastructure — placing them beyond the reach of undergraduate research teams.
- **Simpler models** are computationally feasible but may miss complex regulatory patterns.
- **Both categories** often lack interpretability: they predict regulatory activity without explaining which sequence features drive the prediction.

This project addresses the gap between predictive power and interpretability at feasible compute scale. We aim to build a Kaggle-friendly, interpretable ML system that predicts regulatory behavior in non-coding DNA and explains which sequence patterns contribute to that prediction — enabling candidate "regulatory switch" discovery without requiring industrial-scale infrastructure.

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2. Vision

> Build the most accessible, interpretable, and reproducible research pipeline for discovering regulatory switch-like patterns in non-coding DNA — starting from public data, running on student-accessible compute, and evolving into a research platform that connects sequence-level predictions to biological insight.

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3. Mission

Develop and maintain an interpretable machine learning pipeline that:

1. **Predicts** whether a non-coding genomic region is likely regulatory (enhancer-like, promoter-like, accessible chromatin) or non-regulatory.
2. **Explains** which sequence motifs, k-mers, or learned features drive each prediction.

3. **Runs** entirely within Kaggle notebooks or equivalent environments during the initial build.
4. **Evolves** toward stronger models, richer interpretability, and broader datasets over time.
5. **Documents** everything for reproducibility and knowledge continuity.

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4. Goals and Non-Goals

4.1 Goals

ID	Goal	Measurable outcome
G1	Build a working regulatory vs. non-regulatory classifier for non-coding DNA	AUROC ≥ 0.80 on held-out test set using ENCODE-derived labels
G2	Produce interpretable outputs for every prediction	Feature importance / motif attribution available for each test region
G3	Validate that identified "important" motifs overlap known biology	≥ 50% of top-10 motifs per cell type match known TF binding motifs in JASPAR/HOCOMOCO
G4	Maintain Kaggle-executable pipeline	All core experiments runnable within Kaggle notebook constraints (16 GB RAM, single GPU, ~12 hr session)
G5	Publish reproducible experiment artifacts	All experiments tracked with logged hyperparameters, metrics, and versioned data references
G6	Create a research-ready codebase	Modular code with tests, documentation, and clear contribution guidelines

4.2 Non-Goals (Explicit Exclusions)

ID	Non-Goal	Rationale
NG1	Train a genomic foundation model from scratch	Requires compute far beyond student-accessible infrastructure
NG2	Build a medical diagnostic product	No clinical validation pathway; regulatory and ethical barriers
NG3	Perform wet-lab experimental validation	No lab access; computational prediction only
NG4	Create a universal genome understanding system	Scope is limited to non-coding regulatory prediction
NG5	Build a CRISPR editing platform	Entirely different domain and infrastructure
NG6	Develop a generic "AI for biology" dashboard	Must stay anchored in the specific scientific task

NG7	Process very long genomic sequences (>10 kb) in early phases	Memory and compute limitations on Kaggle
NG8	Make claims of molecular causality	The project identifies candidate regulatory patterns, not proven causal mechanisms

5. Stakeholders

Stakeholder	Role	Needs
Core student team (2–5 CS undergraduates)	Builders, researchers, maintainers	Clear scope, feasible compute, learning opportunities
Faculty advisor / PI	Scientific oversight, publication guidance	Sound methodology, reproducibility, publishable results
Future collaborators (students joining later)	Contributors	Onboarding documentation, clear codebase, protected scope
Computational biology community	Peer reviewers, users of released tools	Reproducible results, honest claims, open data
Open-source community	Potential downstream users	Well-documented code, permissive licensing, clean APIs

6. Long-Term Impact

If successful, this project:

1. **Demonstrates** that meaningful regulatory pattern discovery is achievable without billion-parameter models or institutional GPU clusters.
2. **Produces** an interpretable analysis pipeline reusable by other groups studying non-coding regulation.
3. **Generates** candidate regulatory switch hypotheses testable by experimental collaborators.
4. **Establishes** a research platform extensible to multi-omics, cross-species, or disease-variant analysis.
5. **Publishes** findings in a computational biology venue (target: workshop paper → conference → journal).

7. Project Constraints

7.1 Compute Constraints

> **Critical constraint.** This shapes every technical decision.

Resource	Available	Limit
GPU	Kaggle (1× NVIDIA T4 or P100, 16 GB VRAM)	30 hrs/week GPU quota
RAM	Kaggle kernel: 16 GB (CPU), 13 GB (GPU)	Per-session limit

Disk	Kaggle: ~20 GB scratch + dataset storage	Per-notebook limit
Session duration	~12 hours maximum	Auto-termination
Multi-GPU	Not available on Kaggle free tier	N/A
TPU	Kaggle TPU v3-8 available (limited weekly quota)	Useful for specific workloads but not default

Implication: Large transformer-based genomic models demand substantial computational resources. Models such as Enformer (249M parameters, 196 kb input), DNABERT-2, Nucleotide Transformer, and HyenaDNA were trained on multi-GPU/TPU setups with significant infrastructure. Training these from scratch is out of scope. Inference or limited fine-tuning of smaller pretrained checkpoints may be explored in later phases only.

7.2 Data Constraints

- Public datasets only (no proprietary or restricted-access data in initial phases).
- ENCODE, GEO, and UCSC Genome Browser are primary sources.
- Data must be downloadable, parseable, and processable within Kaggle storage limits.
- Curated subsets preferred over full-database downloads.

7.3 Team Constraints

- 2–5 undergraduate CS students.
- Strong in software engineering and ML; variable in computational biology background.
- No dedicated bioinformatician on the team (biology knowledge will be acquired through literature).
- Asynchronous work expected; no guaranteed full-time commitment.

7.4 Ethical Constraints

- No human-subjects data requiring IRB approval.
- ENCODE data is publicly available under consortium data use policies.
- No personally identifiable genomic information is used.
- All model claims must be framed as computational predictions, not clinical assertions.

7.5 Timeline Constraints

- Initial working baseline: within 3 months.
- Research-grade results: within 6 months.
- Publication-ready work: within 12 months.
- The project is designed for incremental progress, not a single deliverable deadline.

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8. Success Criteria (Summary)

Timeframe	Definition of success
1 month	Data pipeline functional; at least one baseline model trained and evaluated

3 months	Interpretable baseline with motif analysis operational; reproducible results
6 months	Multi-model comparison complete; interpretability report generated; draft research write-up
12 months	Platform extensible; potential publication submitted; documented for handoff

Detailed milestones are in Document 10 — Roadmap & Milestones.

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9. Governance

- **Scope changes** require documented justification and team consensus (see Glossary — Scope Lock).
- **Model upgrades** beyond lightweight architectures must pass decision gates defined in Modeling Roadmap.
- **Data additions** must be reviewed for licensing, size, and relevance before integration.
- **Claims** in any report or publication must distinguish between computational prediction and biological causality.

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10. Scope Lock Statement

This section is canonical. It must be preserved and referenced in all project documents.

Dimension	Locked scope
Domain	Non-coding regulatory genomics
Core objective	Discover and interpret candidate regulatory switch-like patterns in non-coding DNA
Early execution environment	Kaggle-friendly
Early model philosophy	Interpretable and lightweight first
Large genomic transformer training	Out of scope for initial build (compute limits)
Preferred starting data	Public datasets such as ENCODE
Long-term evolution	Stronger models, richer interpretation, broader generalization — only after a working baseline exists

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End of Document 01 — Project Charter Next: 02 — Product Requirements Document